Geophys. J. R. astr. Soc. (1978) **54**, 669–680

Palaeomagnetism of the Nexø Sandstone from Bornholm Island, Denmark

S. N. Prasad* and P. V. Sharma *Institute of Geophysics, University of Copenhagen, 2200 Copenhagen, N. Denmark*

Received 1978 February 15; in original form 1977 December 12

Summary. Samples from the Nexø Sandstone of the Lower Cambrian–Precambrian boundary in South Bornholm reveal a stable NRM with a direction after magnetic cleaning of $D = 226°$, $I = -30°$ ($\alpha_{95} = 11.5°$). This NRM appears to originate in the detrital hematite grains rather than in the red cement of the sandstone. The stable NRM is likely to be of primary origin and reflects a Lower Cambrian pole at 104° W, 38° N (dp = 7°, dm = 11°). Apparent discrepancies between the Bornholm pole and the few other published Early Cambrian/Late Precambrian poles from the Baltic Shield are consistent with the suggestion of large polar movements in those times.

1 Introduction

Deviating Lower Palaeozoic poles have been reported for the Cambrian (Briden, Irons & Johnson 1971; Poorter 1972a; Storetvedt 1973; Tarling 1974) as well as for the Ordovician rocks (Creer 1970; Briden, Morris & Piper 1973; Piper 1974; Thomas & Briden 1976). Most of the authors concerned tend to interpret the discrepant results as revealing a series of rapid polar shifts during the Lower Palaeozoic, but Storetvedt (1973) attributes the deviating results to false palaeomagnetic directions 'resulting from oversimplified analysis of complex magnetic components', and he continues to believe that the Lower Palaeozoic pole was apparently fixed for at least 300 Myr.

A well-defined Lower Cambrian palaeomagnetic polar path is of great importance, firstly for continuation of the polar curve from the Early Palaeozoic to the Late Precambrian times, and secondly for a study of the polar pattern at around the time of the Eocambrian glaciation. For this study the samples were collected from the Nexø sandstone of Bornholm because this hematite-bearing rock promised to be usable for palaeomagnetic investigation. The derived Lower Cambrian pole from the present study provides another deviating result from North Europe.

* Present address: National Geophysical Research Institute, Hyderabad, 500007, India.

Downloaded from https://academic.oup.com/gji/article/54/3/669/612337 by Curtin University Library user on 26 May 2026

2 Geologic setting and age

The Danish island of Bornholm, covering an area of about 600 km^2, is situated in the Baltic Sea south of Sweden. The island is a fault-bounded horst of the Fennoscandian border zone and its geology is closely related to the geology of South Sweden. Reviews of the geology are given by Milthers (1930), Noe-Nygaard (1963) and Gry, Jørgart & Poulsen (1969).

The Precambrian basement in southern and southeastern Bornholm is covered by Palaeozoic and Mesozoic sediments. The Nexø sandstone (henceforth abbreviated as N.S.), consisting of about 100 m of undistributed sequence, is considered to have been deposited in Eocambrian or Early Cambrian times directly on the Precambrian basement rocks of Bornholm (Fig. 1). The N.S. is succeeded by about 60 m of Lower Cambrian marine quartzitic sandstone (Balka sandstone) and by about 100 m of glauconitic siltstone (Gry 1960). The other younger sequences comprise Ordovician and Silurian sediments which are mostly shales. No deposits of Devonian, Carboniferous or Permian age occur in Bornholm.

No direct radiometric or palaeontological dating of the N.S. has so far been possible. The Eocambrian age of the N.S. is favoured on the following considerations:

(1) The formations immediately overlying the N.S. and Balka sandstone are of Lower Cambrian age.
(2) From litho-stratification evidence (Bruun-Petersen 1971) a more or less continuous sedimentation is indicated from N.S. to Balka sandstone.
(3) Based on the correlation of quartz sandstones between definitely Precambrian and Cambrian beds in Scandinavia and Greenland, Poulsen (1977) argues an Early Cambrian stratigraphic age for the N.S.

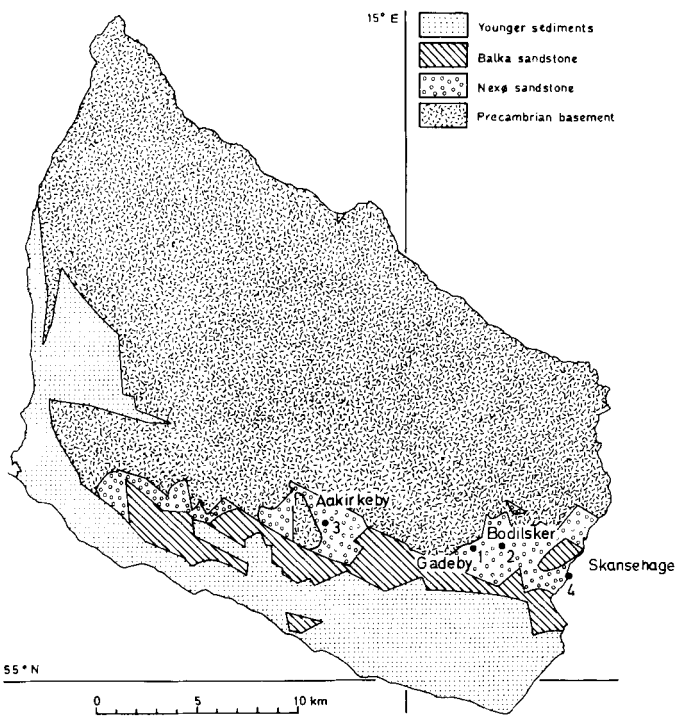


Figure 1. Geological map of Bornholm (after Gry 1960). Sampling localities are shown by numbers.

Downloaded from https://academic.oup.com/gji/article/54/3/669/612337 by Curtin University Library user on 26 May 2026

3 Magnetic mineralogy

The N.S. is a subarkosic sandstone with grains of quartz and feldspar in a red coloured matrix. The lower N.S. sequence exposed at the Gadeby and Bodilsker quarries (Fig. 1) is brownish red and shows alternating dark brown-red and light reddish laminae. The colour difference reflects different amounts of matrix, the dark brown-red being richer in matrix than the light reddish laminae.

The opaque minerals present ($\lesssim 1$ per cent) have a grain size smaller than that of the quartz and feldspar. From microscopic studies (Jensen 1977) it is seen that most of the opaque grains belong to one of the two groups: (1) ilmenite-hematite exsolution intergrowths with the hematite phase dominating, (2) martite with fine lamellar structure originating from titanomagnetite. The relative amount of the group 2 is only about half of that of group 1 in the total amount of opaques in rock specimens from the lower N.S., and is found to be even smaller in specimens from the upper N.S. exposed near Aakirkeby.

The fine lamellar martite could have originated from the titanomagnetites of basement rocks in Bornholm as well as from the Precambrian dolerites in Sweden. The ilmenite–hematite exsolution intergrowths are, however, not known from any basement rock on Bornholm, the nearest exposure of rocks carrying ilmenite–hematite exsolutions lying in Sweden about 150 km northwest of Bornholm. This led Jensen (1977) to suggest that at least part of the material in the N.S. originated in southern Sweden, a proposal that is consistent with the palaeocurrent evidence for the sandstone (Bruun-Petersen 1971).

Mössbauer analysis was made as part of the studies of the magnetic minerals present in N.S. This was essential because the content of opaque minerals was too small to make any tangible assessment based on microscopic studies. When only the powdered fractions of samples were studied (before heavy liquid separation) it took a fairly long time (7 day run) to observe the six-line peaks showing the presence of α-Fe_2O_3. Subsequently, the magnetic concentrates (density > 2900 kg/m^3) obtained after heavy liquid separation were studied for

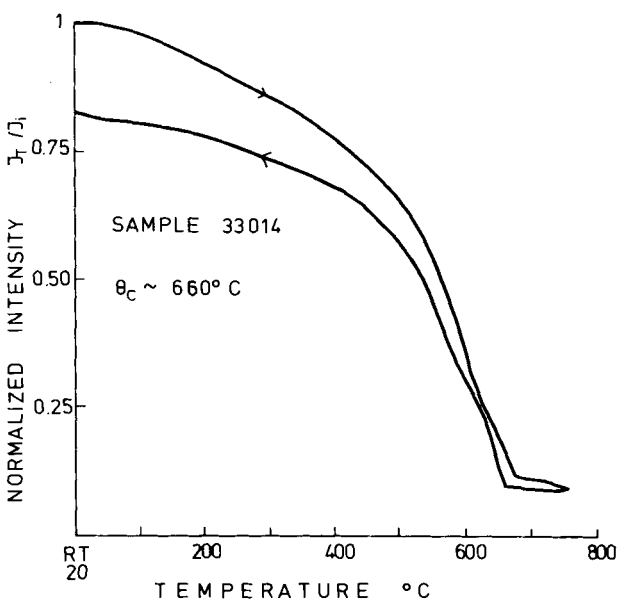


Figure 2. Curie temperature curve for a Nexø sandstone sample. Intensity normalized with respect to induced magnetization J_i in a field of 500 000 A/m (≈ 6.3 kOe).

Downloaded from https://academic.oup.com/gji/article/54/3/669/612337 by Curtin University Library user on 26 May 2026

Mössbauer analysis which indicated a distinct presence of α-Fe_2O_3. Besides hematite, no other phase of magnetic mineral could be detected. Red coating separate of powdered samples did not give any detectable Mössbauer line spectra. The hematite phase thus appeared to be the principal carrier of magnetization.

Curie temperature studies on a few samples were made in the Palaeomagnetic laboratory at Bergen, Norway. The separated magnetic fractions were subjected to a translational torque by a magnetizing field and the intensities plotted as a function of temperature. A typical curve, representative for the samples studied, is shown in Fig. 2. Here again the only recognizable magnetic mineral phase is that of hematite.

4 Sampling and measurement procedures

In all, 78 hand samples were collected from 21 sites at four localities (Fig. 1) where the N.S. sequence of about 30 m is exposed mainly in quarries. An average of four hand samples per site were collected, these being taken from the same horizon and a few metres apart wherever possible. The vertical spacing between the sampling sites varied from 1 to 2 m. The N.S. beds show a gentle dip of 2–5° and it was necessary to apply dip correction only in a few cases.

NRM of the specimens was measured on a PAR SM-1 spinner magnetometer in the Copenhagen laboratory. NRM intensities ranged from 4.8×10^{-2} to 1.6×10^{-4} A/m for a large majority (~ 70 per cent) of specimens. The specimens from six sites at localities no. 3 and 4 (Fig. 1) were too weak for measurement by the spinner in the laboratory environments at Copenhagen.

Low-field susceptibility values ranged from 1.2×10^{-4}–2×10^{-5} SI. On a few pilot specimens, measurements for directional susceptibility were also made to test for anisotropy. These measurements, which were made with a Czech Kappameter in use at the Bergen laboratory, showed the anisotropy in all cases to be < 12 per cent, which is not capable of deflecting the direction of magnetization by more than a few degrees (Fuller 1963).

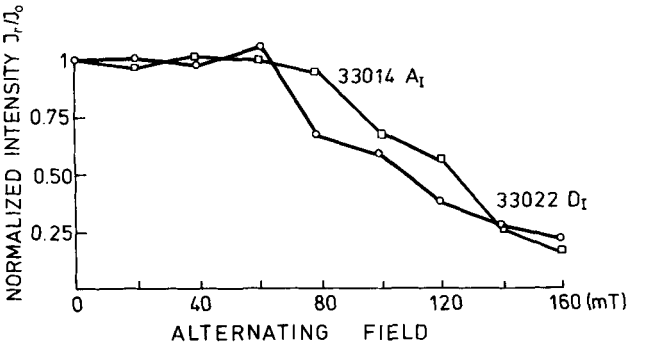
5 Stability and origin of the NRM

5.1 AF DEMAGNETIZATION

To check the stability of NRM, on average two specimens from each site were subjected to af demagnetization in increasing peak fields from 10 mT ($1T = 10^9 \gamma$) onwards. No appreciable changes in intensity and direction were observed up to peak fields of 50 mT. The changes in the NRM intensity during af demagnetization for some representative specimens are shown in Fig. 3(a). For a great majority of specimens the median destructive field ranged between 100 and 130 mT, indicating high coercivity of the NRM.

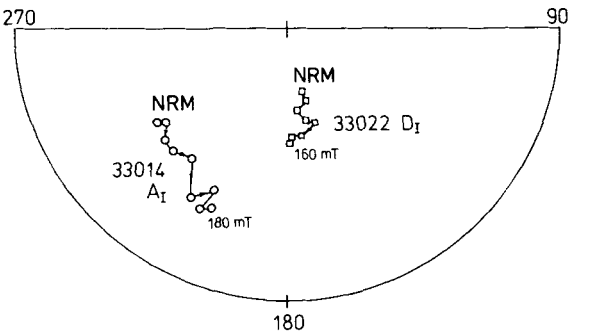
The directional behaviour of NRM with progressive af demagnetization was found to be inconsistent in that a few specimens attained stable vectors at about 80 mT whereas for others the stabilization of the remanence vectors needed peak fields as high as 180 mT (Fig. 3(b)). This necessitated selective af treatment in varying peak fields from 80 to 180 mT for batch cleaning of specimens from all sites. The af demagnetization unit at Copenhagen has been frequently tested for ARM with various orientations of a specimen in the tumbler with increasing peak fields up to 200 mT. The repeat measurements with N.S. specimens showed the ARM to be insignificant at this level.

Downloaded from https://academic.oup.com/gji/article/54/3/669/612337 by Curtin University Library user on 26 May 2026



(a)

Figure 3. (a) af demagnetization curves for two Nexφ sandstone specimens. Intensity normalized with respect to the initial NRM (J_0).



(b)

Figure 3. (b) Change in directions of NRM of the two specimens (Fig. 3(a)) with progressive af demagnetization. Remanent directions are plotted on a polar equal area net.

With progressive af demagnetization the vectors showed a general tendency to move towards shallower inclinations (Fig. 3(b)), however, within-site scatter in directions even after treatment at the 160 mT peak value is rather large. A possible explanation for this scatter could be that the af cleaned data may still include some secondary components of high coercivity. Four sites (G_2, G_3, B3, B6) showed no directional consistency between samples after af treatment, and were eliminated from the group analysis. The mean remanent directions, statistical parameters and other relevant information are shown in Table 1.

5.2 THERMAL DEMAGNETIZATION

Thermal demagnetization of a few specimens from each site was carried out to test their stability and possibly also to ascertain the cause of large scatter in the af cleaned remanent vectors. The large scatter in the remanent vectors could possibly originate from a secondary magnetization component, e.g. PTRM or TCRM. A stepwise thermal demagnetization was carried out in a dynamically controlled magnetic environment and the remanent vectors measured by a DIGICO spinner at the Bergen laboratory.

Thermal demagnetization curves for specimens from three sites are shown in Fig. 5(a), all suggesting the presence of a dominant component of magnetization with a blocking temperature around 600°C, presumably carried by hematite–ilmenite intergrowths. The local drop in intensity at about 500°C observed in some specimens, may be indicative of the

Downloaded from https://academic.oup.com/gji/article/54/3/669/612337 by Curtin University Library user on 26 May 2026

Table 1. Palaeomagnetic direction and pole position derived for the Nexø sandstone (Eocambrian).

Locality/site	N (af)	R	Remanent direction after af cleaning $D°$	$I°$	K	α_{95}	N (td)	R	Remanent direction after thermal cleaning $D°$	$I°$	K	α_{95}
Gadeby – G1	9	8.54	240	−42	18	13	6	5.55	303	−46	11	21
G2	8	5.33	–	–	–	–	6	5.86	235	−44	36	11
G3	6	3.21	–	–	–	–	6	5.28	222	−14	7	26
G4	14	11.88	195	−31	6	18	4	3.86	205	−22	21	20
G5	14	12.47	151	−26	9	14	4	3.90	225	−20	32	16
G7	9	8.33	191	−35	12	15	6	5.76	206	−48	21	15
G8	11	10.80	228	−26	50	8	5	4.94	204	−20	66	9
G9	6	5.93	253	−19	72	8	6	5.64	236	−31	14	18
G10	6	5.89	200	−31	46	10	6	5.91	256	−12	53	9
G11	11	8.84	202	−32	5	23	5	4.76	212	−17	17	19
Bodilsker – B3	8	4.67	–	–	–	–	6	5.92	202	−25	59	8
B6	7	4.23	–	–	–	–	6	5.88	215	20	41	10
B8	11	10.41	223	−26	17	11	5	4.87	201	−28	30	14
B9	12	10.84	216	−30	10	15	6	5.44	239	−40	9	24
Mean for above sites		(9.14)	210	−33	11	16		(13.08)	226	−30	13	11.5
Corresponding pole position			121° W	47° N ($dp = 11°$, $dm = 20°$)					104° W	38° N ($dp = 7°$, $dm = 11°$)		

Notes:

N(af) = number of specimens af cleaned.
N(td) = number of specimens thermally cleaned.
R = resultant of N unit vectors.
K = precision parameter after Fisher (1953).
α_{95} = radius of circle of confidence at the 95 per cent probability level.
dp, dm = radii of oval of confidence of pole positions.

Downloaded from https://academic.oup.com/gji/article/54/3/669/612337 by Curtin University Library user on 26 May 2026

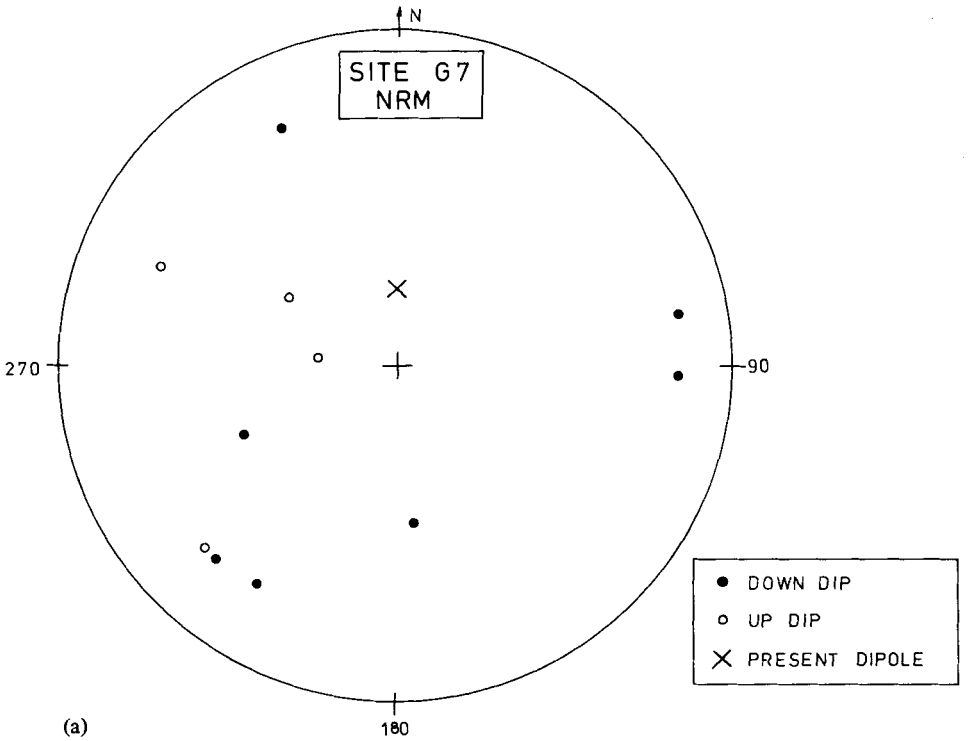


Figure 4. (a) NRM directions plot on a polar equal area net for one sampling site of Nexφ sandstone.

presence of a significant amount of titanomagnetite. The thermal demagnetization studies are thus consistent with the magnetic mineralogy of the N.S. (Section 3). Any possibility of magnetic mineralogical changes having occurred during heating of the sandstone was checked by measuring the bulk susceptibility of specimens before and after heating. Within the accuracy of measurements the repeat measurements showed no appreciable changes in susceptibility of specimens.

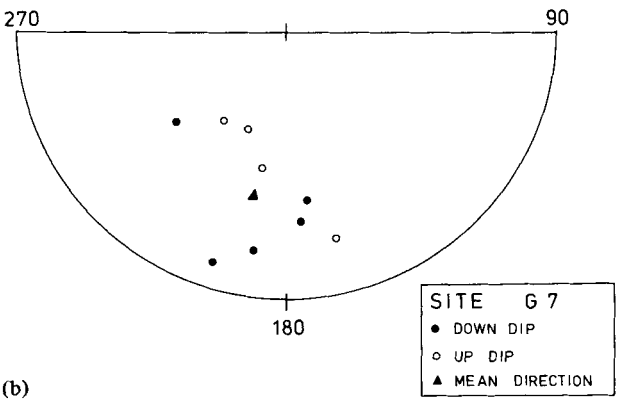


Figure 4. (b) Remanent magnetic directions from the same site (Fig. 4(a)) after af cleaning shown on a polar equal area net. Two specimens were rejected because they became very weak after af treatment in a peak field of 160 mT.

Downloaded from https://academic.oup.com/gji/article/54/3/669/612337 by Curtin University Library user on 26 May 2026

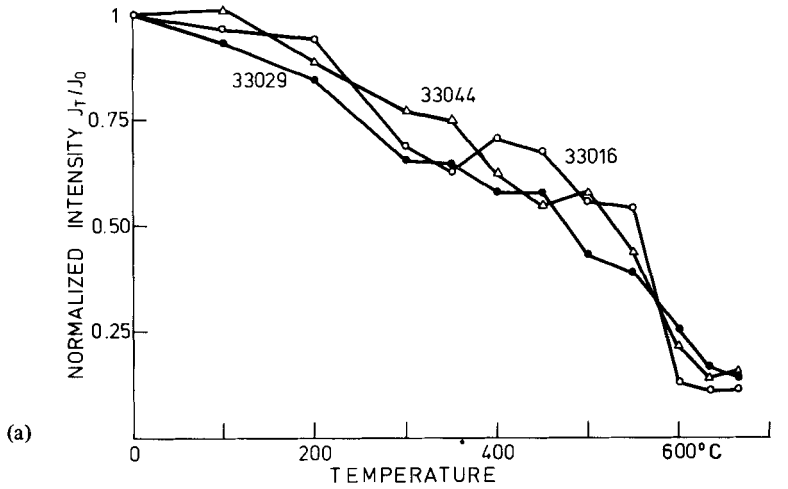


Figure 5. (a) Thermal demagnetization curves for three Nexø sandstone specimens. Intensity normalized with respect to the initial NRM (J_0).

A few representative examples of directional changes with progressive thermal demagnetization are shown in Fig. 5(b). Heating up to 200°C did not appreciably change the remanent vectors. Significant directional changes started occurring at temperatures above 300°C and varied from specimen to specimen. Their analysis does not show any well-defined directional migration so as to identify a secondary component of magnetization with distinct

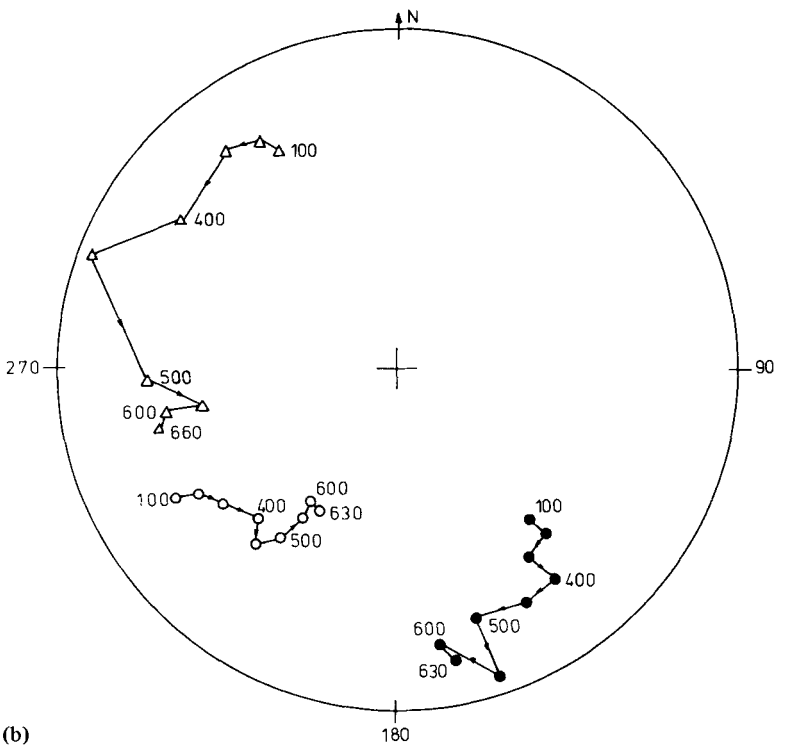


Figure 5. (b) Change in directions of NRM of the three specimens (Fig. 5(a)) with progressive thermal demagnetization shown on a polar equal area net.

Downloaded from https://academic.oup.com/gji/article/54/3/669/612337 by Curtin University Library user on 26 May 2026

blocking temperatures, the only general tendency being for the directions to move towards shallower inclinations in the southwest quadrant. In all cases stable directional end points were reached at temperatures between 600 and 660°C. The statistics of calculation of all site mean directions are summarized in Table 1. The site means have been computed with unit weight to specimens. It is worth mentioning that a large number of sites showed a better directional consistency within thermally cleaned samples than in the af cleaned samples, though the mean direction obtained from the thermally cleaned data (14 sites) is not significantly different at 95 per cent probability level from that obtained using the af cleaned data (10 sites).

5.3 CHEMICAL TREATMENT

Chemical demagnetization was tried on a few pilot specimens to study the relative contributions of the red coloured matrix and the black pigment to the NRM. During initial treatment with a mild solution (10 per cent) no appreciable change either in intensity or direction of the remanence vector was found in the specimens even after five days of treatment, the only noticeable effect was the disappearance of the red coating. Only with increased acid strength (25 per cent) and more than six hours of treatment did the remanent vectors show significant changes in intensity and direction. The directional behaviour of NRM vector with increasing period of treatment (6 to 72 hr) varied from specimen to specimen, indicating no systematic changes to be consistent with those resulting from af and thermal treatment. The likely cause of this might be the varying depth of penetration of the acid from specimen to specimen, depending on the content of the cementing material. Treatment with phosphoric acid gave similar results.

Whatever its cause, the chemical demagnetization was not effective in yielding stable remanent directions in the pilot specimens. However, the fact that there was no significant change in the NRM intensity with the mild HCl treatment that was effective in removing the red coating, suggests the main carrier of the NRM to be the black pigment.

6 Palaeomagnetic results

The mean remanent directions obtained after af and thermal demagnetization are given separately in Table 1, and the difference between them is not significant at acceptable probability levels. However, on two grounds: (1) the af cleaned data may still include some secondary component of high coercivity and (2) the thermally cleaned data are available for a larger number of sites, we consider the mean direction obtained from the thermally cleaned data ($D = 226°$, $I = -30°$, $\alpha_{95} = 11.5°$) to be more reliable for the primary magnetization of the N.S.

No definite explanation can yet be given for a rather poor grouping of the remanent vectors at many sites even after af or thermal cleaning. As mentioned earlier, the studies of directional susceptibility rule out any significant anisotropy effect in the N.S. specimens. However, there is always the possibility that a rock sample may be isotropic and yet strongly inhomogeneous as to the distribution of magnetization resulting, in part, due to uneven spread of the magnetic minerals in the rock matrix. This might have occurred at various stages during the diagenetic history of the N.S. Alternatively, rather large variations in the direction and intensity of the ambient field during the slow deposition/consolidation of the magnetic grains could make the remanent magnetization strongly inhomogeneous in the bulk of the samples. The spinner measurements made with various spins of a specimen showed, in particular, an unusually large divergence in the values of horizontal components of remanent

Downloaded from https://academic.oup.com/gji/article/54/3/669/612337 by Curtin University Library user on 26 May 2026

magnetization, thus indicating a marked inhomogeneity which was encountered in the bulk of the N.S. specimens. Which of the above possibilities has played a major role in disharmonizing the remanent magnetization in the bulk of N.S. samples is a subject for further investigation.

7 Implications of the results

The virtual geomagnetic pole for Bornholm computed from the stable remanent direction in the N.S. is located at 104° W, 38° N. This pole is widely separated from other Lower Cambrian poles (Fig. 6) derived from contemporary formations in Norway and British Isles.

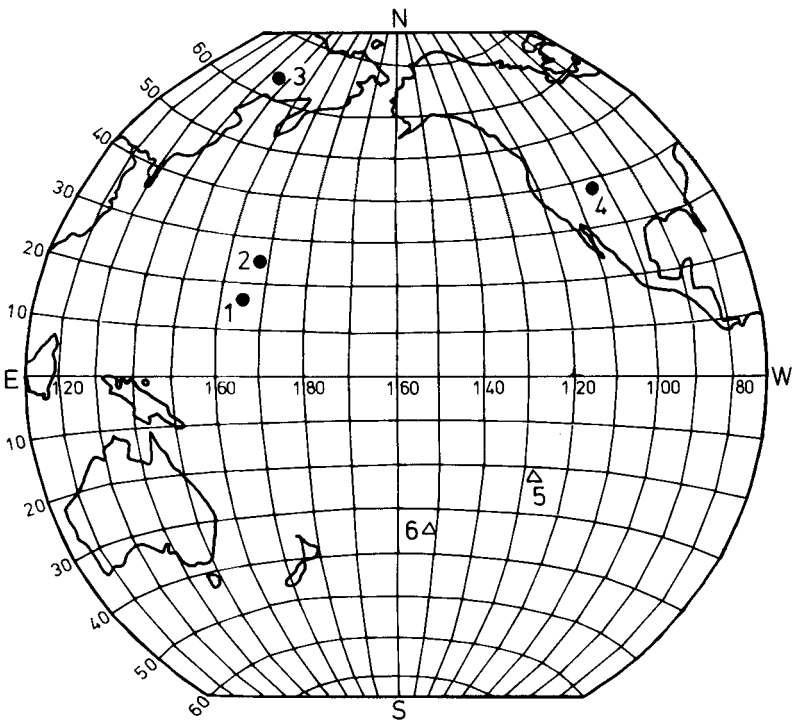


Figure 6. Lower Cambrian and Late Precambrian pole positions of Western Europe. (Only the results based on af or thermal cleaned data included. Late Precambrian poles are only from the Baltic shield.) (1) Hartshill Quartzite, Lower Cambrian (dp = 10°, dm = 18°, Bidgood 1961), (2) Caerfai series, Lower Cambrian (dp = 7°, dm = 13°, Briden, Irons & Johnson 1971), (3) Fen rødberg, Eocambrian (dp = 3°, dm = 4°, Poorter 1972a), (4) Nexø sandstone, Eocambrian (dp = 7°, dm = 11°, this study), (5) Egersund dolerite, 663 Myr (dp = 14°, dm = 16°, Poorter 1972b), (6) Hunnedalen dykes, Late Precambrian (dp = 9°, dm = 10°, Poorter 1972b).

It may be mentioned that discrepancies of comparable magnitude also occur in other Lower Palaeozoic pole positions from Norway and Britain, and the simplest interpretation (Briden 1970; Briden *et al.* 1973) proposed for the deviating poles is that these are derived from different crustal plates (the Baltic/Russian plate and the British Isles subplate). The above interpretation may as well explain the divergence between the Lower Cambrian poles derived for Bornholm (pole 4) and the British Isles (poles 1 and 2). However, to account for the large deviation between the Bornholm pole and the Fen rødberg pole from South Norway (pole 3), both derived from the same crustal plate, we have to resort to some other explanation.

Downloaded from https://academic.oup.com/gji/article/54/3/669/612337 by Curtin University Library user on 26 May 2026

One possibility might be to speculate for N.S. an age much older than Lower Cambrian. From the available stratigraphic control, the N.S. sequence lies definitely between Late Precambrian and Lower Cambrian. Although from lithostratigraphy and other supporting evidence mentioned earlier (Section 2), an Eocambrian age was strongly favoured, an older age in Late Precambrian could not be totally excluded. However, from the plot of poles (Fig. 6) there appears to be no correlation between the Bornholm pole and the Late Precambrian poles from adjacent areas of the Baltic shield (poles 5 and 6). Also, since the Bornholm pole does not correspond to any later (Permian to Ordovician) known position for poles from stable Europe, any possibility of remagnetization of N.S. in the Late- or Mid-Palaeozoic appears to be rather remote for the present.

The Eocambrian poles derived for the Nexø sandstone and the Fen rødberg show an apparent polar separation of about 56° of arc. Unless the magnetization ages of the two standstone formations are much different, the large divergence in their VGP's is probably a reflection of rapid polar shifts that might have occurred at the time of acquisition of their magnetizations. This is not improbable in the light of considerable evidence from elsewhere in support of large polar movements at the time of Late Precambrian glaciations (McElhinny, Giddings & Embleton 1974).

Acknowledgments

We are thankful to P. Mark and J. Schuldt for assistance in collection of samples, and to F. Träff and O. Kristensen for assistance with laboratory instruments. We have greatly benefited from our discussions with J. Bruun-Petersen and Aage Jensen of the Geological Institute, University of Copenhagen.

Thanks are also due to the staff members of the Geophysics Department, University of Bergen for providing facilities to conduct part of the reported investigations at the Bergen laboratory. We are also thankful to the anonymous referees for their suggestions which led to an improvement of the paper.

A grateful acknowledgment is made to DANIDA for the grant of a fellowship to one of us (SNP) during the tenure of the project.

References

Bidgood, D. E. T., 1961. Differential secondary magnetization in some British Cambrian rocks, *Nature*, **192**, 39–40.

Briden, J. C., 1970. Palaeomagnetic results from the Arrochar and Garabal Hill–Glen Fyne Igneous Complexes, Scotland, *Geophys. J. R. astr. Soc.*, **21**, 457–470.

Briden, J. C., Irons, J. & Johnson, P. A., 1971. Palaeomagnetic studies of the Caerfai series and the Skomer Volcanic Group (Lower Palaeozoic, Wales), *Geophys. J. R. astr. Soc.*, **22**, 1–16.

Briden, J. C., Morris, W. A. & Piper, J. D. A., 1973. Palaeomagnetic studies in the British Caledonides–VI. Regional and global implications, *Geophys. J. R. astr. Soc.*, **34**, 107–34.

Bruun-Petersen, J., 1971. Nexø-sandstenens petrografi & lithostratigrafi, *unpublished thesis* (in Danish), University of Copenhagen.

Creer, K. M., 1970. A review of palaeomagnetism, *Earth Sci. Rev.*, **6**, 369–466.

Fisher, R. A., 1953. Dispersion on a sphere, *Proc. R. Soc. London A*, **217**, 295–305.

Fuller, M. D., 1963. Magnetic anisotropy and palaeomagnetism, *J. geophys. Res.*, **68**, 293–309.

Gry, H., 1960. *Geology of Bornholm*, 21 Int. Geol. Congr. Guide A 45-C 40, 16 pages.

Gry, H., Jørgart, T. & Poulsen, V., 1969. *Geology of Bornholm (Geologi på Bornholm)*, VARV Ekskursionsfører Nr. 1, 64 pages.

Jensen, Aa., 1977. Character and provenance of the opaque minerals in the Nexø sandstone, Bornholm, *Bull. geol. Soc. Denmark*, **26**, 69–76.

Downloaded from https://academic.oup.com/gji/article/54/3/669/612337 by Curtin University Library user on 26 May 2026

McElhinny, M. W., Giddings, J. W. & Embleton, B. J. J., 1974. Palaeomagnetic results and late Precambrian glaciations, *Nature Phys. Sci.*, **248**, 557–61.

Milthers, V., 1930. *Bornholms geology*, D.G.U., V. Raekke, **1**, 140 pages.

Noe-Nygaard, A., 1963. *The Precambrian of Denmark*, ed. Rankama, K., pp. 1–26 (The Precambrian, Vol. **1**), Interscience Publishers.

Piper, J. D. A., 1974. Sulitjelma Gabbro, Northern Norway: a palaeomagnetic result, *Earth planet. Sci. Lett.*, **21**, 383–388.

Poorter, R. P. E., 1972a. Preliminary palaeomagnetic results from the Fen carbonatite complex, S. Norway, *Earth planet. Sci. Lett.*, **17**, 194–198.

Poorter, R. P. E., 1972b. Palaeomagnetism of the Rogaland Precambrian (Southwestern Norway), *Phys. Earth planet. Int.*, **5**, 167–176.

Poulsen, V., 1978. The Precambrian–Cambrian boundary in parts of Scandinavia and Greenland, *Geol. Mag.*, in press.

Storetvedt, K. M., 1973. Implications from fossil magnetism of the Fen complex (Cambrian), Southern Norway, *Phys. Earth Planet. Int.*, **7**, 450–460.

Tarling, D. H., 1974. A palaeomagnetic study of Eocambrian tillites in Scotland, *J. geol. Soc. Lond.*, **130**, 163–177.

Thomas, C. & Briden, J. C., 1976. Anomalous geomagnetic field during the late Ordovician, *Nature Phys. Sci.*, **259**, 380–382.

Downloaded from https://academic.oup.com/gji/article/54/3/669/612337 by Curtin University Library user on 26 May 2026